

Milad Tatar Mamaghani

Sydney, NSW, Australia

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🏠 Homepage • 📄 Google Scholar • 🔗 LinkedIn • 📄 ORCID • 🐙 GitHub

EDUCATION

Doctor of Philosophy (Ph.D.) in Engineering

MONASH UNIVERSITY - CLAYTON CAMPUS

Jan. 2019–Dec. 2022

Melbourne, Australia

- Thesis: *Safeguarding Beyond-5G Wireless UAV Communications: Design and Optimization*

Bachelor of Science (B.Sc.) in Electrical–Communications Engineering

AMIRKABIR UNIVERSITY OF TECHNOLOGY (TEHRAN POLYTECHNIC)

Sep. 2012–Oct. 2016

Tehran, Iran

- Thesis: *Secure Communications with Untrusted Relaying and Wireless Energy Harvesting*
- Also B.Sc. in Control Engineering with additional course credits (Feb. 2018)

RESEARCH EXPERIENCE

ARC Postdoctoral Research Fellow (Level B)

THE AUSTRALIAN NATIONAL UNIVERSITY — SCHOOL OF ENGINEERING

Jan. 2023–Jan. 2025

Canberra, Australia

- Pursued research on physical-layer security, short-packet communications, and integrated sensing and communications.
- Published 3 first-author papers in Q1 IEEE transactions (1x IEEE JSAC, IF 17.2, 2x IEEE TWC IF 10.4)
- Presented 2 papers at flagship conferences (1x IEEE ICC 2024 and 1x IEEE Globecom 2023).
- Received 2 best-paper awards (IEEE SPCC-TC BPA 2024; IEEE ICC BPA 2024 — top 0.6% of 2,364 submissions).
- Co-supervised 1 PhD and 2 R&D engineering students, supporting research and educational activities.

Subject Matter Expert — AI Research Consultant

MERCOR / SCALE AI (REMOTE)

Feb. 2025–Present

Sydney, Australia

- Drive research-based projects through technical evaluation and domain-specific model reasoning for top-tier AI labs.
- Benchmark AI models for complex algorithmic reasoning and mathematical accuracy in STEM contexts.
- Curate high-fidelity training datasets to improve AI alignment, factual precision, and reasoning.
- Formulate advanced prompting strategies and evaluation guidelines for next-generation AI models.

Doctoral Researcher — Wireless Communications

MONASH UNIVERSITY — DEPARTMENT OF ELECTRICAL AND COMPUTER SYSTEMS ENGINEERING

Jan. 2019–Dec. 2022

Melbourne, Australia

- Pursued research on wireless communications security, resource allocation, signal processing, and IoT network design.
- Designed reinforcement learning and mathematical optimization frameworks for energy-efficient airborne communications.
- Developed aerial MIMO beamforming and energy-efficient trajectory planning architectures to secure UAV networks.
- Published 7 peer-reviewed articles in leading journals; received 5+ awards, grants, and accolades for research excellence.

Research Assistant

AMIRKABIR UNIVERSITY OF TECHNOLOGY — WIRELESS COMMUNICATION RESEARCH LAB

Jan. 2016–Nov. 2018

Tehran, Iran

- Researched physical-layer security, untrusted relay networks, wireless energy harvesting, and cooperative jamming.
- Produced 1 conference paper and 2 journal articles during early research and development.

Research Intern

IRAN TELECOMMUNICATION RESEARCH CENTER (ITRC) — FIXED COMMUNICATIONS GROUP

Jun. 2016–Sep. 2016

Tehran, Iran

- Worked on IoT systems; gained hands-on experience on applied telecommunications systems and network infrastructure.

RESEARCH INTERESTS

My research develops theoretical and algorithmic foundations for intelligent wireless systems that are secure, reliable, and efficient under strict latency and resource constraints. I focus on understanding and exploiting fundamental trade-offs between communication, sensing, and learning in complex wireless environments. My work integrates information theory, optimisation, signal processing, and machine learning to design next-generation 6G networks and autonomous wireless infrastructures. Current research directions include:

- AI-Native Wireless Networks and Learning-Driven Resource Allocation
- Machine Learning and Reinforcement Learning for Wireless Communications
- Physical-Layer Security, Privacy, and Trustworthy Network Design
- Ultra-Reliable Low-Latency Communications (URLLC)
- Integrated Sensing and Communications (ISAC)
- Emerging 6G Technologies: UAV, THz, RIS, etc.

HONOURS & RECOGNITIONS

- 2025 **IEEE ComSoc Best Readings in Covert Communication and Networking [Link]**
- Recognises impactful and landmark technical contributions in the field. Featured paper [9].
- 2024 **IEEE ICC Best Paper Award [Certification]**
- Selected as top 0.6% of 2,364 submissions across all symposia.
- 2024 **IEEE Signal Processing and Computing for Communications Best Paper Award [Certification]**
- Recognises outstanding contributions to the field by the Technical Committee; only one paper is awarded each year.
- 2024 **Australian Global Talent Recognition**
- Recognises individuals of exceptional achievement in technology under the national innovation program.
- 2022 **Exceptional Ph.D. Thesis**
- Recognised among the top 5% of Ph.D. theses in the field by external examiners.
- 2019 **Vice Chancellor's Congratulatory Letter for Teaching Excellence**
- Awarded for outstanding performance in teaching and learning based on student evaluations at Monash University.
- 2014 **Exceptional Talent**
- Recognised for outstanding academic excellence during undergraduate studies at the Amirkabir University of Technology.
- 2012 **Top 0.2%, National University Entrance Examination (Konkour)**
- Ranked in the top 0.2% among nearly 300,000 applicants nationwide.
- 2011 **Semifinalist, National Mathematics and Chemistry Olympiads**
- Selected among top performers in national mathematics and chemistry competitions.

AWARDS & GRANTS

- 2023 **ANU Early Career Researcher International Travel Grant (AUD \$3000)**
- An early-career researcher grant awarded by The Australian National University for presenting research at international conferences.
- 2022 **Monash Postgraduate Publications Award (AUD \$5000)**
- Recognises for outstanding research publications during doctoral candidature.
- 2022 **Monash Graduate Research Completion Award**
- Awarded for successful and timely completion of doctoral research.
- 2022 **Monash Engineering Faculty's Postgraduate Research Scholarship**
- Recognises outstanding research potential and academic achievement in the field of engineering.
- 2019 **Monash Graduate Scholarship and International Tuition Scholarship, Monash University**
- Stipend (AUD \$120,000) and full tuition waiver (AUD \$200,000) over the course of doctoral studies.

TEACHING EXPERIENCE

Teaching Associate

Feb. 2019–Dec. 2022

MONASH UNIVERSITY — FACULTY OF ENGINEERING

Melbourne, Australia

- Facilitated hands-on learning through effective tutoring, practical sessions, and lab demonstrations.
- Provided mentorship that enhanced student performance in coursework and engineering design projects.
- Contributed to curriculum delivery by designing assessments and offering constructive feedback to students.
- Mentored students, assisted in designing assessments, marking, and evaluating presentations.
- Achieved an overall student satisfaction score of 4.73/5.0, ranking in the top 7.3% of all University units in the semester.

Courses Taught

- ECE5179: Neural Networks and Deep Learning; Tutor and Lab Demonstrator, Semester 2, 2022
- ECE5883: Advanced Signal Processing; Lab Designer and Demonstrator, Semester 1, 2022
- ENG5001: Advanced Engineering Data Analysis; Tutor, Semester 1, 2022
- FIT1048: Fundamentals of C++; Tutor and Lab Demonstrator, Semester 1, 2022
- ECE5884: Wireless Communications; Lab Demonstrator, Semester 2, 2021
- ECE4191: Engineering Design; Lab Demonstrator and Supervisor, Semester 2, 2020–2022
- ENG5105: Integrated Design; Coordinator and Project Mentor, Semester 2, 2020–2022
- ECE4146: Multimedia Technologies; Lab Demonstrator, Semester 2, 2020–2022
- ECE3141: Information and Networks; Lab Demonstrator, Semester 1, 2020–2022
- ECE4044: Telecommunications Protocols; Lab Demonstrator, Semester 1, 2020
- ECE2072: Digital Systems; Tutor and Lab Demonstrator, Semester 1, 2020–2022
- ECE4042: Communications Theory; Tutor and Lab Demonstrator, Semester 1, 2019–2022
- ECE4076: Computer Vision; Tutor, Lab Demonstrator, and Exam Coordinator, Semester 1, 2019–2022
- ECE2071: Computer Organisation and Programming; Tutor and Lab Demonstrator, Semester 1, 2019–2022

STUDENT SUPERVISION

Yu Li

2023–Present

PH.D. CANDIDATE, THE AUSTRALIAN NATIONAL UNIVERSITY

- **Topic:** Physical-Layer Anonymous Communications; co-supervised with A/Prof. Xiangyun Zhou.

Mac Gilligan

2024–2025

ENGINEERING R&D HONOURS STUDENT, THE AUSTRALIAN NATIONAL UNIVERSITY

- **Topic:** SDR implementation for over-the-air secure file transmission with GNU Radio.

Reetika

2023–2024

VISITING BACHELOR STUDENT, INDIAN INSTITUTE OF SPACE SCIENCE AND TECHNOLOGY

- **Topic:** Physical-layer security for terahertz communications; co-supervised with Prof. Nan Yang.

Engineering Design Project Teams

2021–2022

UNDERGRADUATE AND POSTGRADUATE CAPSTONE PROJECTS, MONASH UNIVERSITY

- **Automated Robotic Arm:** Color-cube sorting robotic arm with microprocessor-based controller, 2021.
- **Autonomous Mobile Robot:** Odometric sensing for high-level navigation and autonomous ball collection, 2022.
- **Intelligent Baby Monitor:** Sustainable monitor with voice/image recognition and real-time parent alerts, 2022.

PROFESSIONAL SERVICE & LEADERSHIP

Conference Organisation and Selection Service

- **Technical Program Committee Member** — IEEE Vehicular Technology Conference, 2024.
- **Session Chair** — IEEE Global Communications Conference, Physical Layer Security Symposium, 2023.
- **Selection Committee Member** — ANU Summer Research Scholarship Program, 2023.
- **Exam Supervisor** — Monash University Online Assessment & Student Services, 2021–2023.

Peer Review Contributions

- **Volunteer Technical Reviewer** for 25+ international journals and conferences with 140+ verified reviews, 2018–2026.
 - **Journals:** IEEE JSAC, TWC, TCOM, TIFS, TSP, TVT, TCCN, TMC, TGCN, TNSE, IoT-J, Systems Journal, Access, Communications Letters, Wireless Communications Letters, Communications Magazine, Wireless Communications Magazine, Network Magazine, IET Communications, Physical Communication, Security and Communication Networks, Vehicular Communications, and Engineering Computations.
 - **Conferences:** IEEE ICC (2020–2023), IEEE GLOBECOM (2017, 2024), IEEE VTC (2024), WiOpt (2023), WCSP (2023), CyberC (2020), and IEEE ICC Workshop on Integrating UAVs into 5G and Beyond (2020).

Professional Memberships

- IEEE Member (2022–2026); IEEE Graduate Student Member (2020–2022).
- IEEE Communications Society (2020–2025); IEEE Signal Processing Society (2020–2022).
- IEEE Young Professionals (2020–2025); IEEE Consultants Network (2020–2025).
- IEEE Quantum Community (2020–2022); IEEE Computer Society (2020); IEEE Big Data Community (2020).

SELECTED TALKS & PRESENTATIONS

- On the Information Leakage Performance of Secure Finite Blocklength Transmissions over Rayleigh Fading Channels, presented at the *Communication Theory Symposium: Physical Layer Security and Privacy* session, IEEE ICC 2024, Denver, CO, USA, June 2024. [Slides][Certificate]
- On the Secrecy Performance of Finite Blocklength Communications over Fading Channels, presented at the *21st Australian Communications Theory Workshop (AusCTW)*, Melbourne, Feb. 2024. [Poster]
- Secure Short-Packet Transmission with Aerial Relaying: Blocklength and Trajectory Co-Design, presented at the *Main Symposium on Communication and Information System Security (CISS)* session at IEEE Globecom 2023, Kuala Lumpur, Malaysia, Dec. 2023. [Slides][Certificate]
- Final Review Seminar (Ph.D. Thesis Defense): Safeguarding Wireless Communications with Unmanned Aerial Vehicles: Design and Optimization, presented at the ECSE Department, Monash University, Melbourne, Australia, Feb. 2022. [Slides]

PUBLICATIONS

- **Scholarly outputs:** 12 journal articles, 3 conference papers, 2 preprints, and 2 theses
- **Research Impact:** 672 citations, h-index 12, i10-index 12 (as of June 2026)

Journals

- [12] **Milad Tatar Mamaghani**, Xiangyun Zhou, Nan Yang, and Lee Swindlehurst, “Securing integrated sensing and communication against a mobile adversary: A Stackelberg game with deep reinforcement learning,” *IEEE Journal on Selected Areas in Communications*, vol. 44, pp. 942–958, Sep. 2025. (IF 17.2, **Q1**)
DOI 10.1109/JSAC.2025.3611404 |  PDF |  arXiv |  Code
- [11] **Milad Tatar Mamaghani**, Xiangyun Zhou, Nan Yang, Lee Swindlehurst, and Vincent Poor, “Performance analysis of finite blocklength transmissions over wiretap fading channels: An average information leakage perspective,” *IEEE Transactions on Wireless Communications*, vol.

- 23, no. 10, pp. 13252–13266, Oct. 2024. (IF 10.4, Q1)
DOI 10.1109/TWC.2024.3400601 |  PDF |  arXiv |  Code
- [10] **Milad Tatar Mamaghani**, Xiangyun Zhou, Nan Yang, and Lee Swindlehurst, “Secure short-packet communications via UAV-enabled mobile relaying: Joint resource optimization and 3D trajectory design,” *IEEE Transactions on Wireless Communications*, vol. 23, no. 7, pp. 7802–7815, July 2024. (IF 10.4, Q1)
DOI 10.1109/TWC.2023.3344802 |  PDF |  arXiv |  Code
- [9] **Milad Tatar Mamaghani** and Yi Hong, “Aerial intelligent reflecting surface-enabled terahertz covert communications in beyond-5G Internet of Things,” *IEEE Internet Things Journal*, vol. 9, no. 19, pp. 19012–19033, March 2022. (IF 9.9, Q1)
DOI 10.1109/JIOT.2022.3163396 |  PDF |  arXiv |  Code
- [8] **Milad Tatar Mamaghani** and Yi Hong, “Terahertz meets untrusted UAV-relaying: Minimum secrecy energy efficiency maximization via trajectory and communication co-design,” *IEEE Transactions on Vehicular Technology*, vol. 71, no. 5, pp. 4991–5006, May 2022. (IF 6.41, Q1)
DOI 10.1109/TVT.2022.3150011 |  PDF |  arXiv
- [7] **Milad Tatar Mamaghani** and Yi Hong, “Joint trajectory and power allocation design for secure artificial noise aided UAV communications,” *IEEE Transactions on Vehicular Technology*, vol. 70, no. 3, pp. 2850–2855, Mar. 2021. (IF 6.41, Q1)
DOI 10.1109/TVT.2021.3057397 |  PDF |  arXiv
- [6] **Milad Tatar Mamaghani** and Yi Hong, “Intelligent trajectory design for secure full-duplex MIMO-UAV relaying against active eavesdroppers: A model-free reinforcement learning approach,” *IEEE Access*, vol. 9, pp. 4447–4465, Dec. 2020.
DOI 10.1109/ACCESS.2020.3048021 |  PDF
- [5] **Milad Tatar Mamaghani** and Yi Hong, “Improving PHY-security of UAV-enabled transmission with wireless energy harvesting: Robust trajectory design and communications resource allocation,” *IEEE Transactions on Vehicular Technology*, vol. 69, no. 8, pp. 8586–8600, May 2020. (IF 6.41, Q1)
DOI 10.1109/TVT.2020.2998060 |  PDF |  arXiv
- [4] **Milad Tatar Mamaghani**, Ali Kuhestani, Hamid Behrouzi, “Can a multi-hop link relying on untrusted amplify-and-forward relays render security?,” *Wireless Networks*, vol. 27, pp. 795–807, Nov. 2020.
DOI 10.1007/s11276-020-02487-w |  PDF |  arXiv
- [3] **Milad Tatar Mamaghani** and Yi Hong, “On the performance of low-altitude UAV-enabled secure AF relaying with cooperative jamming and SWIPT,” *IEEE Access*, vol. 7, pp. 153060–153073, Oct. 2019.
DOI 10.1109/ACCESS.2019.2948384 |  PDF |  Code
- [2] **Milad Tatar Mamaghani** and Robert Abbas, “Security and reliability performance analysis for two-way wireless energy harvesting based untrusted relaying with cooperative jamming,” *IET Communications*, vol. 13, no. 4, pp. 449–459, Mar. 2019.
DOI 10.1049/iet-com.2018.5718 |  PDF
- [1] **Milad Tatar Mamaghani**, Ali Kuhestani, and Kai Kit Wong, “Secure two-way transmission via wireless-powered untrusted relay and external jammer,” *IEEE Transactions on Vehicular Technology*, vol. 67, no. 9, pp. 8451–8465, July 2018. (IF 6.41, Q1)
DOI 10.1109/TVT.2018.2848648 |  PDF |  arXiv

Conferences

- [3] **Milad Tatar Mamaghani**, Xiangyun Zhou, Nan Yang, Lee Swindlehurst, and Vincent Poor, “On the average information leakage of finite blocklength transmissions over Rayleigh fading channels,” in *Proc. IEEE International Conference on Communications*, Denver, CO, USA, pp. 1467–1472, June 2024.
DOI 10.1109/ICC51166.2024.10622223 |  arXiv
- [2] **Milad Tatar Mamaghani**, Xiangyun Zhou, Nan Yang, and Lee Swindlehurst, “Secure short-packet transmission with aerial relaying: Block-length and trajectory co-design,” in *Proc. IEEE Global Communications Conference*, Kuala Lumpur, Malaysia, pp. 5998–6004, Dec. 2023.
DOI 10.1109/GLOBECOM54140.2023.10437972 |  arXiv
- [1] **Milad Tatar Mamaghani**, Abbas Mohammadi, Phil Yeoh, and Ali Kuhestani, “Secure two-way communication via a wireless-powered untrusted relay and friendly jammer,” in *Proc. IEEE Global Communications Conference*, Singapore, pp. 1–6, Dec. 2017.
DOI 10.1109/GLOCOM.2017.8253999 |  arXiv

Under Review

- [2] Azadeh Tabeshnezhad, **Milad Tatar Mamaghani**, Lee Swindlehurst, Tommy Svensson, and Erik Ström, “Robust Beamforming Design for Secure Uplink NOMA-ISAC,” submitted to *IEEE Transactions on Wireless Communications*, Jun. 2026.
DOI 10.48550/arXiv.2606.17306 |  arXiv
- [1] Yu Li, **Milad Tatar Mamaghani**, Xiangyun Zhou, and Nan Yang, “Symbol-Aware Precoder Design for Physical-Layer Anonymous Communications,” submitted to *IEEE Transactions on Communications*, Mar. 2026.
DOI 10.48550/arXiv.2602.20874 |  arXiv

Theses

- [2] **Milad Tatar Mamaghani**, “Safeguarding Beyond-5G Wireless Communications with Unmanned Aerial Vehicles: Design and Optimization,” Ph.D. Thesis, Electrical and Computer Systems Engineering Department, Monash University, Melbourne, Australia, Dec. 2022.
DOI 10.26180/21721709.v1
- [1] **Milad Tatar Mamaghani**, “Secure Communications via Untrusted Relaying with Wireless Energy Harvesting: Performance Analysis and Trade-offs,” B.Sc. Thesis [in Persian], Department of Electrical Engineering, Amirkabir University of Technology, Tehran, Iran, Sep. 2016.